LLM output Configurations:

Output length - Generating more tokens requires more computation from the LLM, leading to higher energy consumption, potentially slower response times, and higher costs.
Reducing output length causes the LLM to stop predicting more tokens once the limit is reached. If your needs require a short output length, you'll also possibly need to engineer your prompt to accommodate.

Sampling Controls-

Temperature: Temperature is a **hyperparameter that controls randomness** in model output.
Model predicts **probabilities for next words**
Temperature **modifies these probabilities before sampling**

Effects of Temperature

Low Temperature (0.0 – 0.3)

- Picks the **highest probability words**
- Output:
    - deterministic
    - repetitive
    - accurate

Used for: code, math, factual answers

Medium Temperature (0.4 – 0.7)

- Balanced output
- Some variation, still reliable

Used for: normal chat, explanations

High Temperature (0.8 – 1.5)

- Flattens probabilities
- More randomness

Output:

- creative
- diverse
- sometimes incorrect

Used for: storytelling, brainstorming

Low temp → **confidence (less randomness)**
High temp → **creativity (more randomness)**

Top-k and Top-p (Nucleus Sampling):

Even with temperature, models can still pick weird low-probability words.
Top-k Sampling: Pick only the top K most probable words, ignore the rest.
Small k → safer, more predictable
Large k → more variety
Top-p (Nucleus Sampling): Pick the smallest set of words whose total probability ≥ p

Effect

- Low p (0.5) → very strict
- High p (0.9–0.95) → more flexible

Temperature → “how random?”
Top-k → “how many choices?”
Top-p → “how much probability mass?”

Real-world defaults

- top_k = 0 (disabled)
- top_p = 0.9 or 0.95
- temperature = 0.7